

Anatomy of the shoulder

Leonardo da Vinci brought both artistic aptitude and scientific enquiry to his studies of anatomy. He often worked in collaboration with an anatomist to ensure the accuracy of his work.

- deltoid
(shoulder muscle)
- biceps brachii
(biceps)
- trapezius
(neck–shoulder muscle)
- scapula (shoulder blade)
connects the humerus
with the clavicle
- clavicle (collarbone)
- humerus (upper arm bone)
- pectoralis (pectoral muscle)
- ribcage
- sternum (breast bone)



1600 BCE

Mummification

In ancient Egypt, bodies are mummified; the internal organs are removed for religious reasons and to help preservation, and are stored in canopic jars.



Canopic jar

12th century

Islamic physicians refute Galenic wisdom

There is no prohibition of human dissection in the medieval Islamic world, where physicians such as Ibn Zuhr (Avenzoar) perform routine autopsies. Ibn Zuhr corrects some of Galen's human anatomy—much of it based on the Barbary ape.

Late 15th century

New observations

With physicians refuting Galen, Leonardo da Vinci begins his study of human anatomy. Italian physician Jacopo da Carpi's *Anatomia Carpi* introduces a new age of original observation.

500 BCE

Early Greek anatomy

The Greek physician Hippocrates promotes animal dissection as a way of learning about the human body.

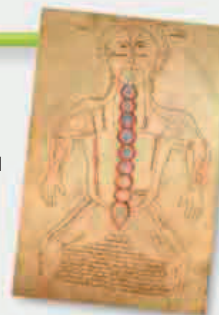


Hippocrates

180 BCE

Galenic circulation

Greek-born physician Galen concludes that blood is continually made in the body, an idea not corrected until the 17th century.



Galen's anatomy

1300s

Mondino de Liuzzi

Italian physician Mondino de Liuzzi performs the first public human dissections c.1315, but his catalog of anatomy perpetuates many erroneous ideas of antiquity.



Liuzzi's *Anatomia*